Evolution and expansion of multidrug-resistant malaria, Greater Mekong Subregion

A regional analysis of whole-genome sequence data was conducted to map the evolution and cross-border spread of multidrug-resistant *P. falciparum* lineages across Cambodia, Laos, north-eastern Thailand, and Vietnam from 2007 to 2018.



Transmission & connectivity analysis

Priority use case: Mapping regional spread and population structure analysis.

Genomic Methodology

Whole-genome sequencing, population structure, resistance-haplotype reconstruction, and lineage mapping (Pf6.2, TRAC2, GenRe-Mekong/SpotMalaria).

KEL1/PLA1 Lineage Prevalence

Spread from Western Cambodia (2009) to >50% prevalence across GMS (excluding Laos) by 2017.

New subgroups with four further *pfcrt* mutations emerging since 2016.

Approach

Pools WGS data from Cambodia, Laos, NE Thailand and Vietnam (2007–2018).

Involves relatedness analysis, resistance-haplotype reconstruction, and lineage spread mapping.

Key findings

Over 1M SNPs analysed. KEL1/PLA1 lineage became dominant across most of GMS, with new drug-resistance mutations emerging in subgroups.

Action supported

Resistance surveillance requires regional data sharing. Genetic relatedness identifies spread; epidemiological data are needed for transmission route inference.

Relevance to this section

Cross-border sharing in practice: pooled national datasets, agreed metadata, and clear attribution across contributing networks.

Full record

Sampling strategy, DNA quality thresholds, align/genotype parameters, mutation frequencies, metadata fields: Case Study 2, Section 13.

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